



Project 1

GROUP C

Dynamic Finite Element Analysis of Structure ME-473

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1 Introduction

The first "mini project" of the course "ME473 - Dynamic finite element analysis of structures" given at EPFL in the spring semester 2025 by Stefano Burzio serves as an introduction to programming finite element calculations in Matlab.

The group analyzes the given problem consisting of a double-clamped uniform bar with given geometrical and material properties, derives the required discretized "*Finite Elements*" equations, and uses them to perform a modal analysis.

1.1 Problem

A uniform bar with length ℓ , modulus E , cross-sectional area A , and density ρ is described, which is clamped at both ends. The problem is considered to be one-dimensional, meaning that only the lengthening in the longitudinal direction of the bar is considered. The natural frequencies for 1, 2, 4, 8, and 16 finite elements are to be calculated and the results compared to the analytical ones, that can be obtained by solving the strong form of the differential problem.

2 Methodology

This mini-project's objective is to solve longitudinal modal frequencies and mode shapes for a double-clamped uniform bar, of isotropic elastic material. This section presents the methodology used to achieve this.

2.1 Problem Definition

Figure 1 presents the reference geometry for the bar examined in this project.

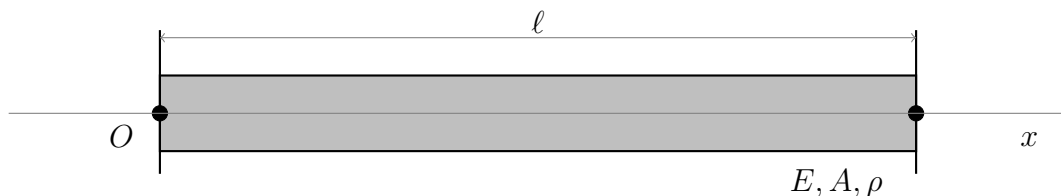


Figure 1: Reference geometry

While geometrical and material properties characterizing the bar are presented in table 1. The values presented are given by the problem statement.

Table 1: Problem's geometrical and material's properties

Property	Symbol	Unit	Value
Length	ℓ	m	1
Section area	A	m ²	0.01
Young's modulus	E	GPa	210
Density	ρ	kg/m ³	7850

2.2 Mechanics

In particular from continuum mechanics, the strong form for the governing equation is derived as:

$$EA\partial_{xx}u_1(x, t) = \rho A\ddot{u}_1(x, t) \quad \forall (x, t) \in]0, \ell[\times]0, T[\quad (1)$$

With the following boundary conditions:

$$u_1(x, t) = \begin{cases} 0, & x = 0, \quad \forall t \in]0, T[\\ 0, & x = \ell, \quad \forall t \in]0, T[\end{cases} \quad (2)$$

Finally, initial conditions are neglected as a modal analysis is conducted.

2.3 Weak Form

Using the virtual work approach on the strong form equations results in:

$$\int_0^\ell \delta u_1 EA \partial_{xx} u_1(x, t) dx = \int_0^\ell \delta u_1 \rho A \ddot{u}_1(x, t) dx \quad (3)$$

Here, a virtual displacement δu_1 is applied to the forces given in the strong form. The resulting virtual work is integrated over the volume, meaning that the equation now relates the virtual work that goes into deformation and inertia. The first term is integrated by parts.

$$\left[\delta u_1 EA \frac{\partial u_1}{\partial x} \right]_0^\ell - \int_0^\ell EA \frac{\partial u_1}{\partial x} \frac{\partial \delta u_1}{\partial x} dx = \int_0^\ell \rho A \ddot{u}_1 \delta u_1 dx \quad (4)$$

Since $\delta u_1 = 0$ on Γ_u the first term vanishes, leaving the definite weak form of the equation:

$$\int_0^\ell EA \frac{\partial u_1}{\partial x} \frac{\partial \delta u_1}{\partial x} dx + \int_0^\ell \rho A \ddot{u}_1 \delta u_1 dx = 0 \quad (5)$$

2.4 Meshing

Since the analysis with different numbers of elements is central to this project, a major focus was placed on having a simple meshing algorithm that could mesh the beam with quadratic elements of equal size.

Having defined N_e as the number of elements in which the beam was divided, $2N_e + 1$ equally spaced nodes have been defined for the elements.

Hence it was required to construct the connectivity table required to assemble the global matrices (Stiffness and Mass) for the existing mesh. Taking into account the convention for 1D quadratic elements, shown in fig 3, it was possible to generalise the connectivity table for a generic element e as follows.

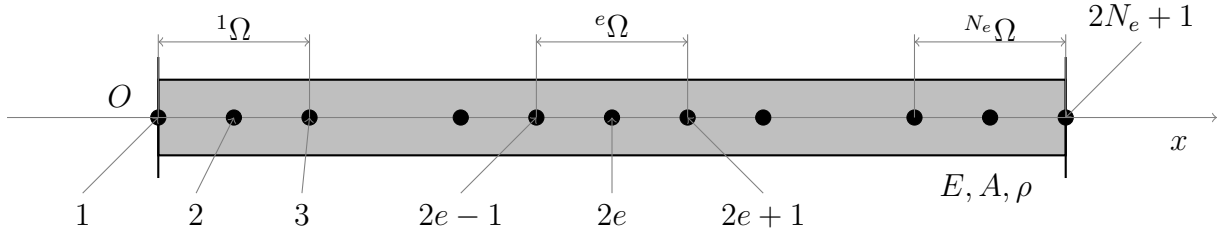


Figure 2: Meshing of the geometrical domain

Nodal id local	Element 1	Element 2	...	Element e	...
Node 1	1	2		$2e - 1$	
Node 2	3	5		$2e + 1$	
Node 3	2	4		$2e$	

Table 2: Connectivity table for the FE mesh

2.5 Finite Elements Equations

The quadratic 1D shape functions $u_1 = \mathbf{H}\vec{q}$, $\delta u_1 = \mathbf{H}\delta\vec{q}$ are defined on the master element as:

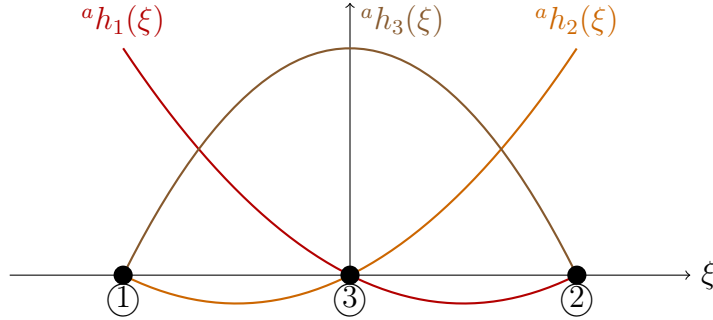


Figure 3: Quadratic Shape Functions for archetypal 1D element

$$\mathbf{H}_a = \left[\xi \frac{\xi - 1}{2}, \xi \frac{\xi + 1}{2}, 1 - \xi^2 \right] \quad (6)$$

Globally, using the shape functions gives the following:

$$\delta\vec{q} \cdot \left(\int_0^\ell EA (\nabla \mathbf{H})^T \nabla \mathbf{H} dx \right) \vec{q} + \delta\vec{q} \cdot \left(\int_0^\ell \rho A \mathbf{H}^T \mathbf{H} dx \right) \vec{q} = 0 \quad (7)$$

The discretization to multiple finite elements, the localization and the use of master elements ($\vec{x}(\xi) = \mathbf{H}_a(\xi) \vec{x}_e$) finally yields:

$$\left(\mathbf{A}_e \int_{\Omega_a} EA (\nabla_\xi (\mathbf{H}_a \mathbf{J}_e^{-1}))^T (\nabla_\xi (\mathbf{H}_a \mathbf{J}_e^{-1})) j_e d\xi \right) \vec{q} + \left(\mathbf{A}_e \int_{\Omega_a} \rho A \mathbf{H}_a^T \mathbf{H}_a j_e d\xi \right) \vec{q} = 0 \quad (8)$$

$$\mathbf{K}\vec{q} + \mathbf{M}\ddot{\vec{q}} = 0 \quad (9)$$

The integrals are thus calculated via the Gauss-Legendre numerical cubic approximation for each element using the master shape functions and the element jacobian matrix ($\mathbf{J}_e = \nabla_{\xi}\vec{x}(\xi)$) and then assembled to the resulting $\mathbf{M}$ and $\mathbf{K}$ matrices. Cubic approximations were chosen to yield exact solutions for quadratic polynomials in 1D. These form an eigenvalue problem, which is then solved to determine the natural frequencies of the studied system.

$$\det(\mathbf{M}\omega^2 - \mathbf{K}) = 0 \quad (10)$$

It is important to point out that boundary conditions are taken into consideration when constructing the global stiffness and mass matrices. Specifically, the rows and columns corresponding to the first and last nodes — those constrained by the boundary conditions — are removed from the $\mathbf{M}$ and $\mathbf{K}$ matrices used in the eigenvalue problem. This adjustment ensures that the constrained nodes are correctly represented as having zero displacements.

3 Results

A finite element modal analysis has been run for five different meshes; meshed with, respectively, 1, 2, 4, 8 and 16 elements. In all instances, elements have been defined equally; that is, with equal length $\ell_e = \ell/N_e$. Table 3, presents the first five natural frequencies obtained for each of the meshes together with the analytically determined natural frequencies, f_j , which cf. table D.2 in [1] are given by

$$f_j = \frac{j}{2} \sqrt{\frac{E}{\rho\ell^2}}, \quad j = 1, 2, \dots \quad (11)$$

While figure 4 shows the first five mode shapes obtained for each of the meshes together with the analytical mode shapes, ϕ_j , which cf. table D.2 in [1] are given by

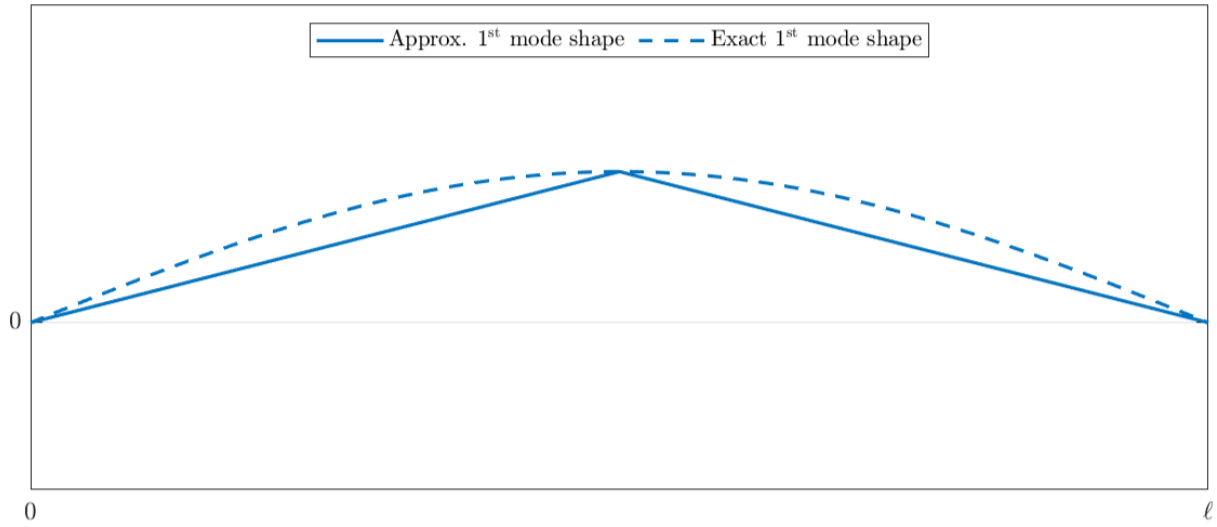
$$\phi_j = \sin\left(j\pi\frac{x}{\ell}\right), \quad j = 1, 2, \dots \quad (12)$$

These mode shapes illustrate the maximum longitudinal displacement from the initial position as the bar oscillates with the corresponding natural frequencies. Upon excitation, the bar

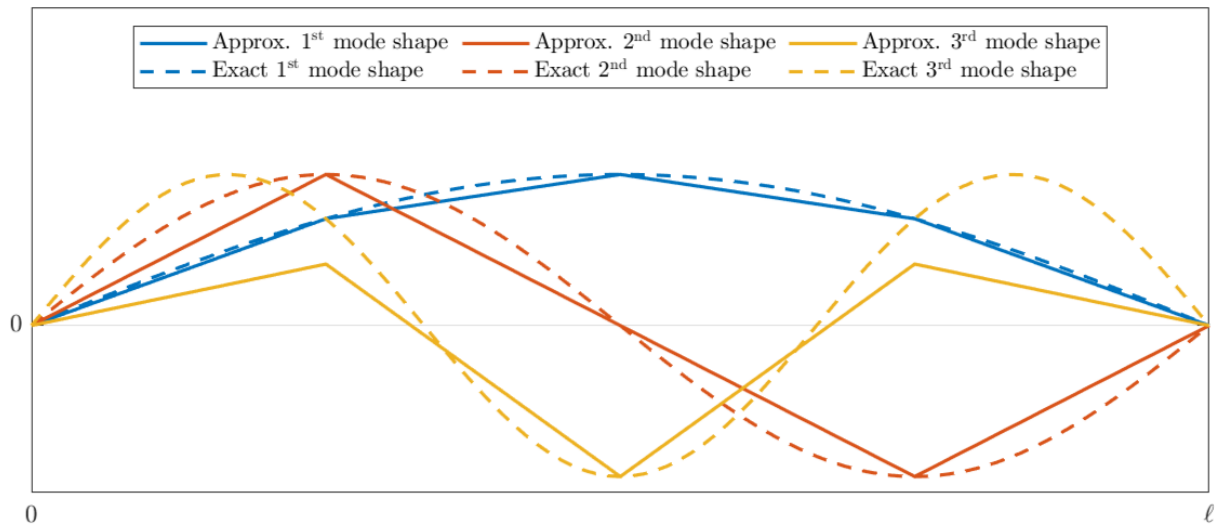
Table 3: Comparison of natural frequencies for the bar obtained using different meshes.

Mode	Analytical frequencies [Hz]	FE frequencies [Hz]				
		$N_e = 1$	$N_e = 2$	$N_e = 4$	$N_e = 8$	$N_e = 16$
1	2586.10	2603.12	2595.81	2586.76	2586.14	2586.10
2	5172.19	—	5206.25	5191.61	5173.52	5172.28
3	7758.29	—	9339.48	7886.42	7757.98	7758.93
4	10344.4	—	—	10412.5	10383.2	10347.0
5	12930.5	—	—	14452.7	13041.9	12938.4

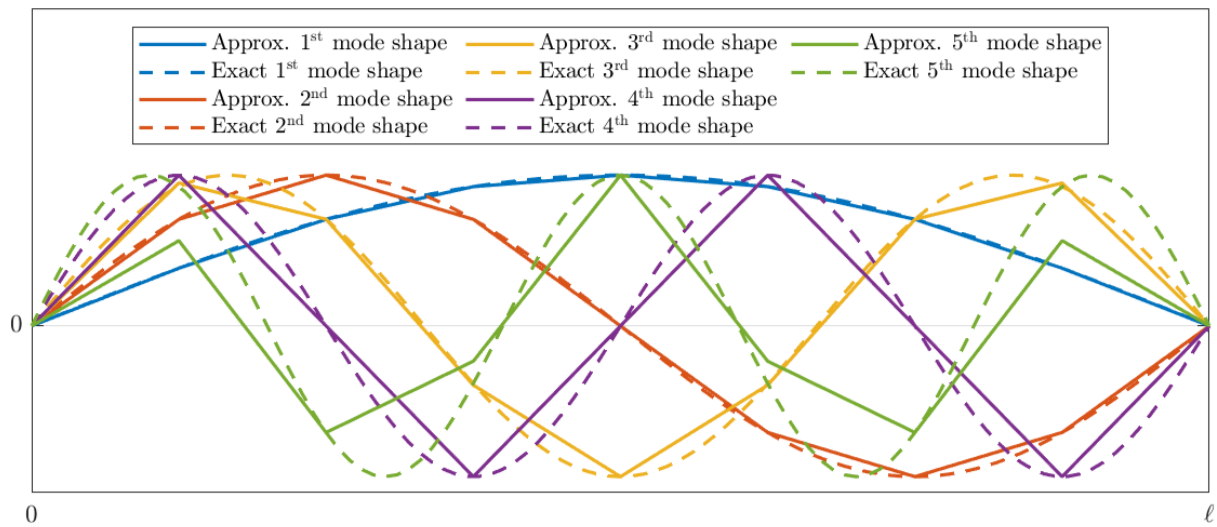
vibrates as a combination of its modes, with the lower modes being the most dominant in terms of amplitude. Therefore, only the five lowest modes are presented.



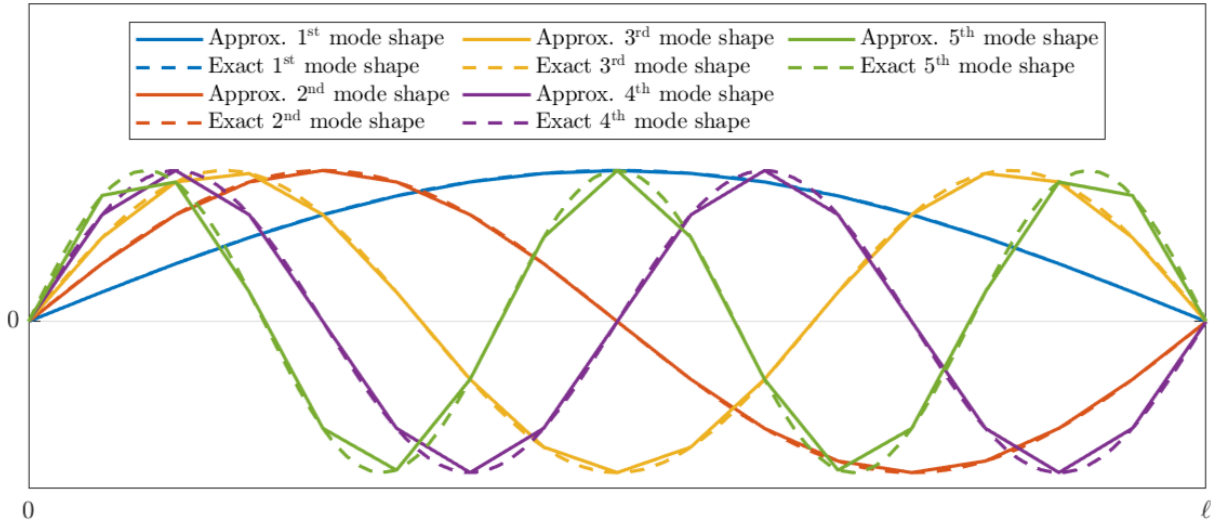
(a) 1 element



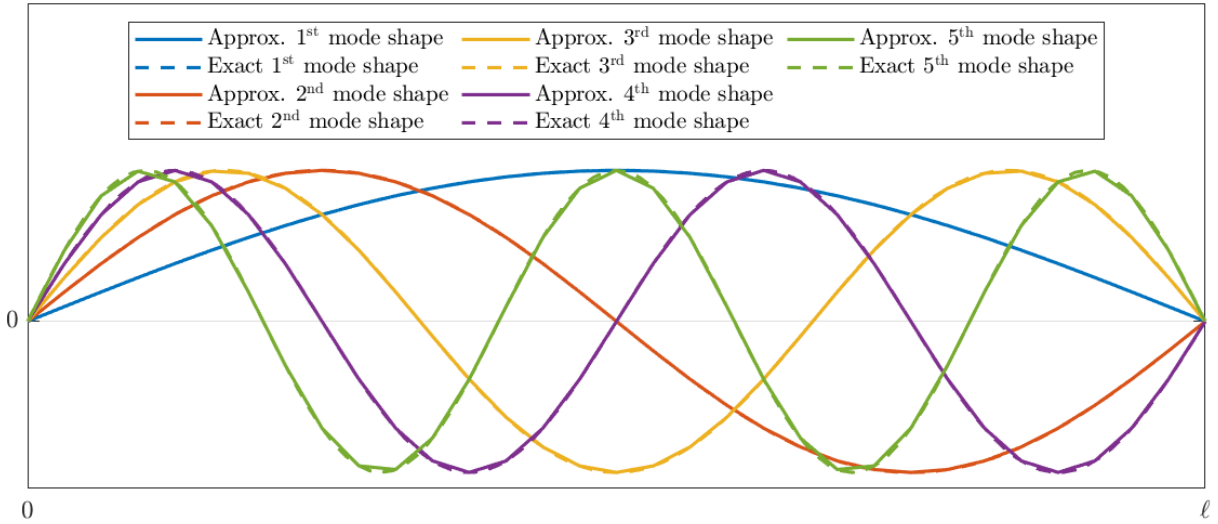
(b) 2 elements



(c) 4 elements



(d) 8 elements



(e) 16 elements

Figure 4: First five mode shapes (when applicable) for different numbers of elements.

To quantify the deviation between analytical frequencies, f_j , and the frequencies estimated with the individual meshes, $\tilde{f}_j$, the relative error ε_j is determined as

$$\varepsilon_j = \left| \frac{f_j - \tilde{f}_j}{f_j} \right| \quad (13)$$

and illustrated in the graph shown in figure 5. The sign of the error is neglected for visual reasons, but one may observe from table 3, that the estimated frequencies are higher than the analytical frequencies in all instances.

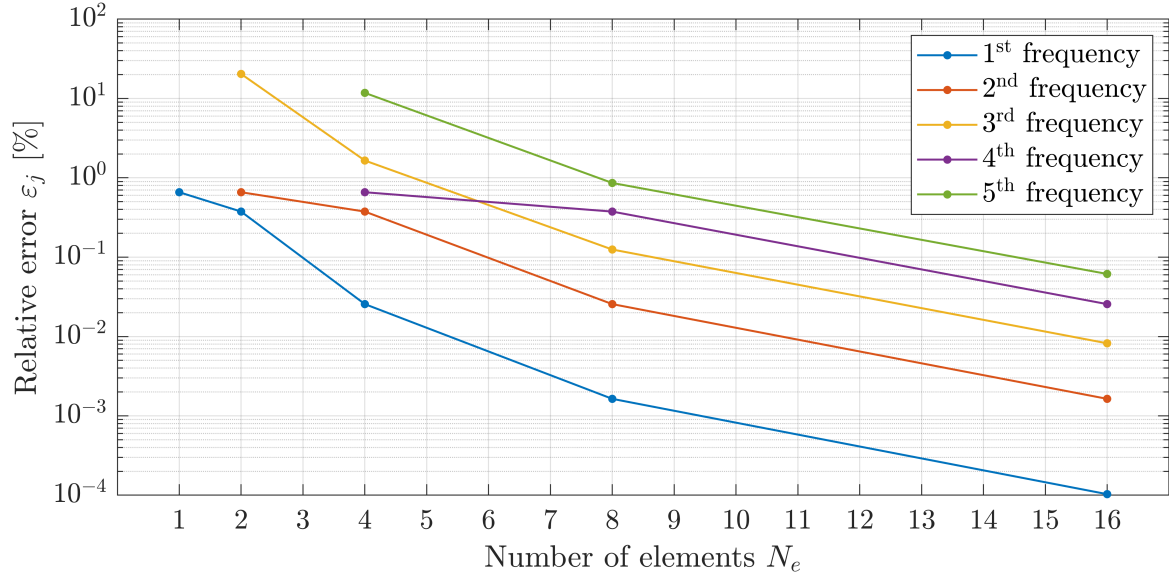


Figure 5: Relative error between analytically determined frequencies and estimated frequencies.

4 Discussion

As this is a simple case, we can observe a very quick convergence towards analytical values, proportional to the number of nodes in each simulation.

The number of eigenfrequencies, as well as the number of eigenvectors and the number of non-zero values in each eigenvector we are able to compute in each simulation is directly equal to the number of free nodes, which itself is equal to the total number of nodes minus the two constrained nodes at each extremity.

The first eigenfrequencies, the most useful in reality, are very quickly (quickly referring to both the number of elements needed, and the computation time) approximated to a relative error inferior to 1%, which is great. Indeed, the first, second and fourth eigenfrequencies are already below this threshold with only 1, 2 and 4 elements respectively (the coarsest mesh used, where they can first be computed, respectively). Regarding the third and fifth eigenfrequencies, it can be observed that they are harder to estimate since they require more iterations to get accurate enough. Nevertheless, no more than 8 elements are required in order to have these eigenfrequencies to have less than 1% relative error. This error drops below 0.1% with 16 elements, which gives a very high accuracy.

We can also observe that increasing the number of elements ($\propto$ number of nodes) increases the accuracy of the eigenfrequencies in ascending order, ie the first eigenfrequency is the first one to get very accurate, followed by the second one and so on.

This extends to the eigenvectors, but with a twist:

For example, with 16 elements, we cannot distinguish the first analytical from the first computed eigenmode (which looks C^1 continuous!), whereas we can still see a discrepancy in the second mode, especially at inflection points, and whereas the fifth mode is still 'jagged'. However, we can also observe that even when 'jagged', the values still tend towards the analytical curve at the nodes, and usually fall on the analytical curve after a few iterations. For instance,

although the first 3 mode shapes can be computed with 2 elements, it can be observed that it is not sufficient to have the three of them to be correct at the nodes. Similarly, the fifth mode shape cannot be correctly computed with 4 elements, thus requiring more elements in the mesh in order to have accurate mode shapes.

The explanation behind the smoother curves for the lower modes is the following: since running the simulation with a certain number of elements will give equally sized eigenvectors (i.e. same number of points in the graph) for each mode, the flatter the mode (the lower its eigenfrequency), the smoother it will appear.

5 Conclusion

In conclusion, we observed realistic, quickly converging results to this 1st Finite Elements simulation in the course.

The best computed values for the first five eigenfrequencies were obtained with the 16 element-simulation:

Mode	FE Frequencies [Hz]	Relative error [-]
1	2586.10	≈ 0
2	5172.28	1.74×10^{-5}
3	7758.93	8.25×10^{-5}
4	10347.0	2.51×10^{-4}
5	12938.4	6.11×10^{-4}

Table 4: Best computed natural frequencies.

We did not encounter any irregularities such as singularity behaviors or locking, which was expected in this case.

One possible design improvement when interested in studying a particular mode shape, would be to have more nodes at the locations of highest magnitudes, for example at $x = \frac{l}{4}$ and $x = \frac{3l}{4}$ for the second mode.

6 References

- [1] Jon Juel Thomsen. *Vibrations and stability*. Vol. 3. Springer, 2021.
- [2] Stefano Burzio. *Dynamic Finite Element Analysis of Structures*. Lecture slides, EPFL, ME-473. Spring Semester 2025.